Name: SOLUTIONS Date Due: ___________

Year 10 Mathematics

Investigation 2, 2015

Topic – Linear Functions

Take home component

AIM: *In this assessment, you will be investigating parallel and perpendicular linear functions.*

Please note: The take home component will be required for the in class component of this investigation. The take home component will be worth **10 bonus marks** on your in class component. If you do not hand in your take home component, you will not be rewarded with these bonus marks.

1. Give definitions for the following words:

a) parallel

Parallel means never meet
Parallel lines are two or more lines that don't intersect.

b) perpendicular

At right angles.

c) gradient

how steep a straight line is

d) linear function

Those whose graph is a straight line.
constant first difference pattern; $y = mx + c$.

e) equation

equation says that two things are equal.
must have an equals sign.

f) y-intercept

The value of y at the point where the line crosses the y axis.

2. For each equation given below, complete the table of values and graph the functions on the axes below.

a) $y = 2x + 1$

x	−2	−1	0	1	2
y	-3	-1	1	3	5

b) $y = 2x - 4$

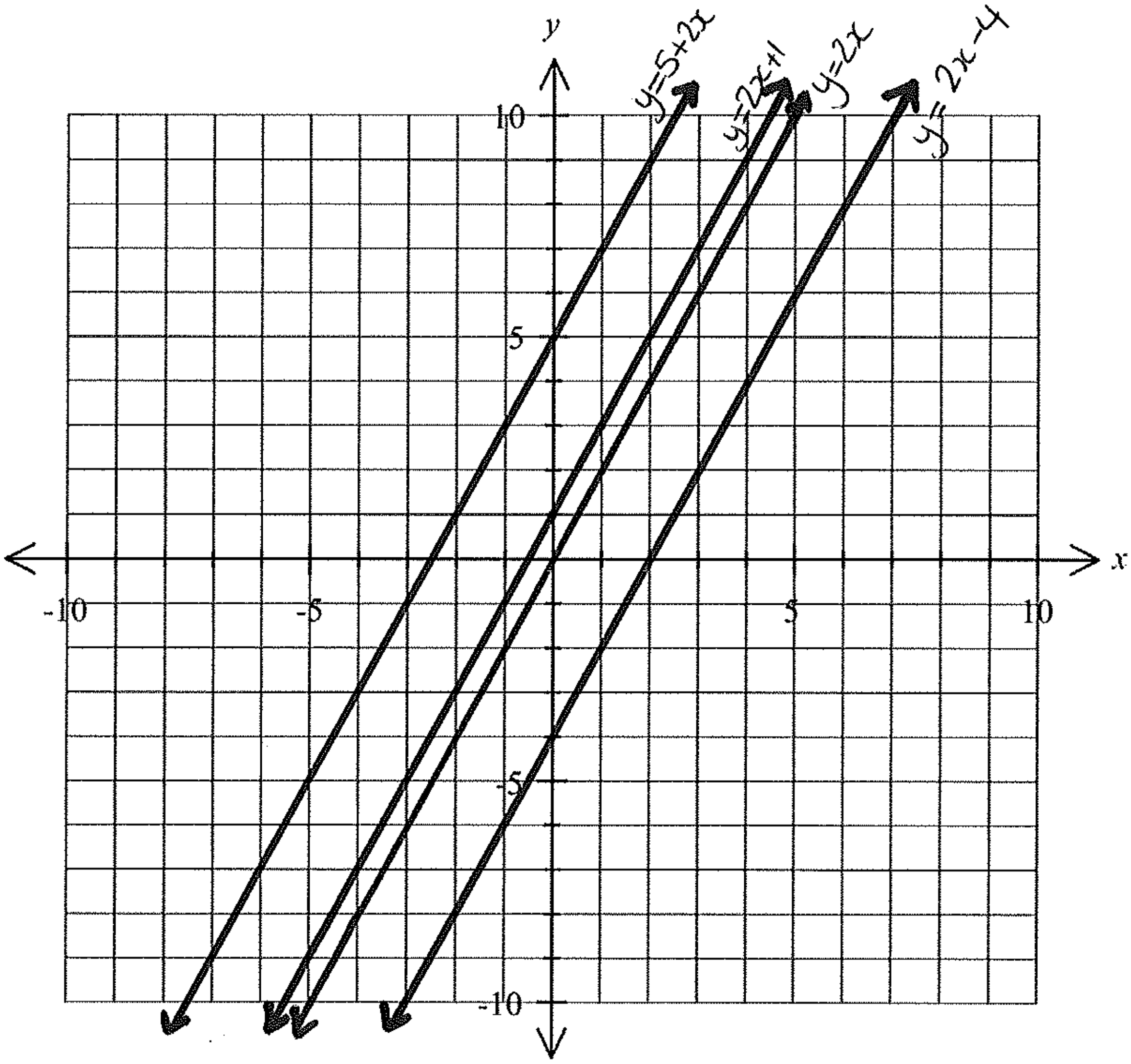
x	−2	−1	0	1	2
y	-8	-6	-4	-2	0

c) $y = 2x$

x	−2	−1	0	1	2
y	-4	-2	0	2	4

d) $y = 5 + 2x$

x	−2	−1	0	1	2
y	1	3	5	7	9

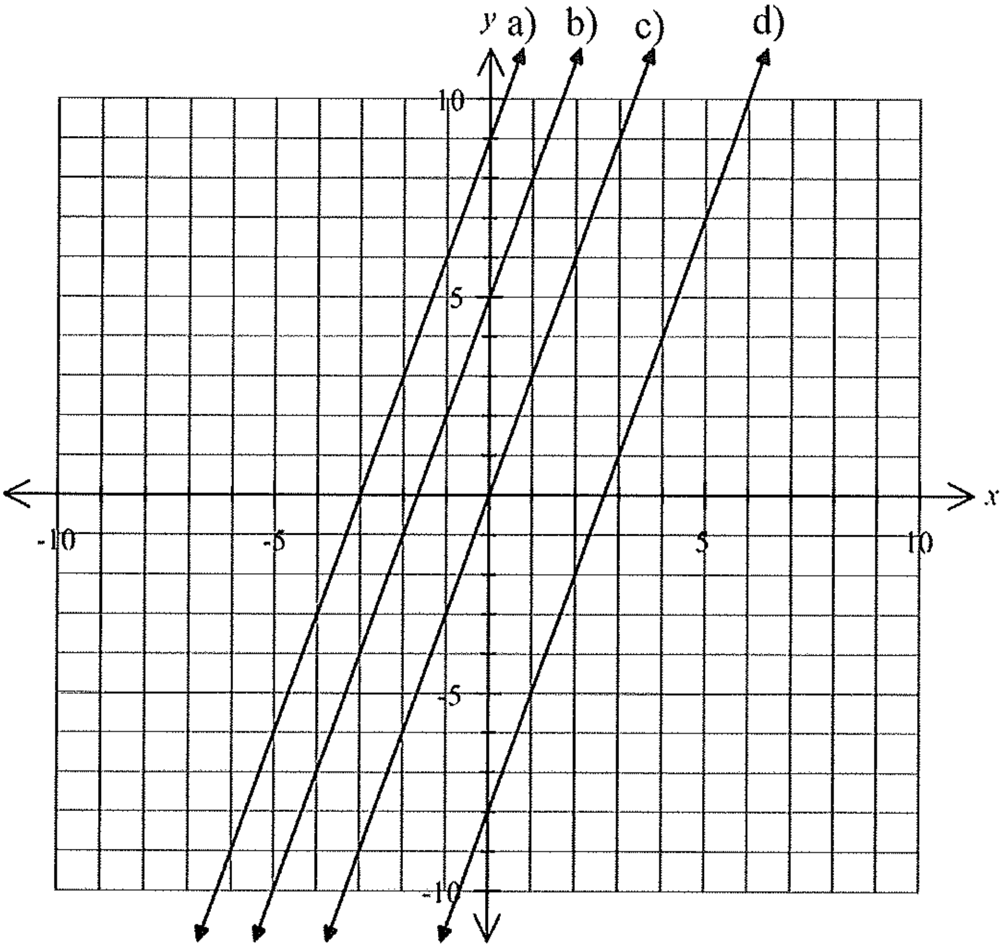


3. What do you notice about the graphs? What is this relationship called?

They don't intersect

Parallel lines.

4. The following axis has graphs drawn on them already. For each graph, complete the table below, stating the gradient, y-intercept and the equation of the linear function.



	Gradient	y-intercept	Equation
a)	3	(0,9)	$y=3x+9$
b)	3	(0,5)	$y=3x+5$
c)	3	(0,0)	$y=3x$
d)	3	(0,-8)	$y=3x-8$

5. What pattern do you notice about the gradients of these lines? What is this relationship called?

The gradient is the same.

Parallel lines.

6. Write a conjecture (guess) about the connection between the gradients of any lines that are parallel.

The gradients of parallel lines are exactly the same.

7. Use your conjecture to state whether each pair of linear functions below are parallel:

a) $y = 5x + 5$ and $y = 5x - 2$ yes

b) $y = 3x - 4$ and $y = -3x + 4$ NO

c) $y = 2x - 9$ and $y = 4x - 9$ NO

d) $y = -7x - 14$ and $y = -7x + 14$ yes.

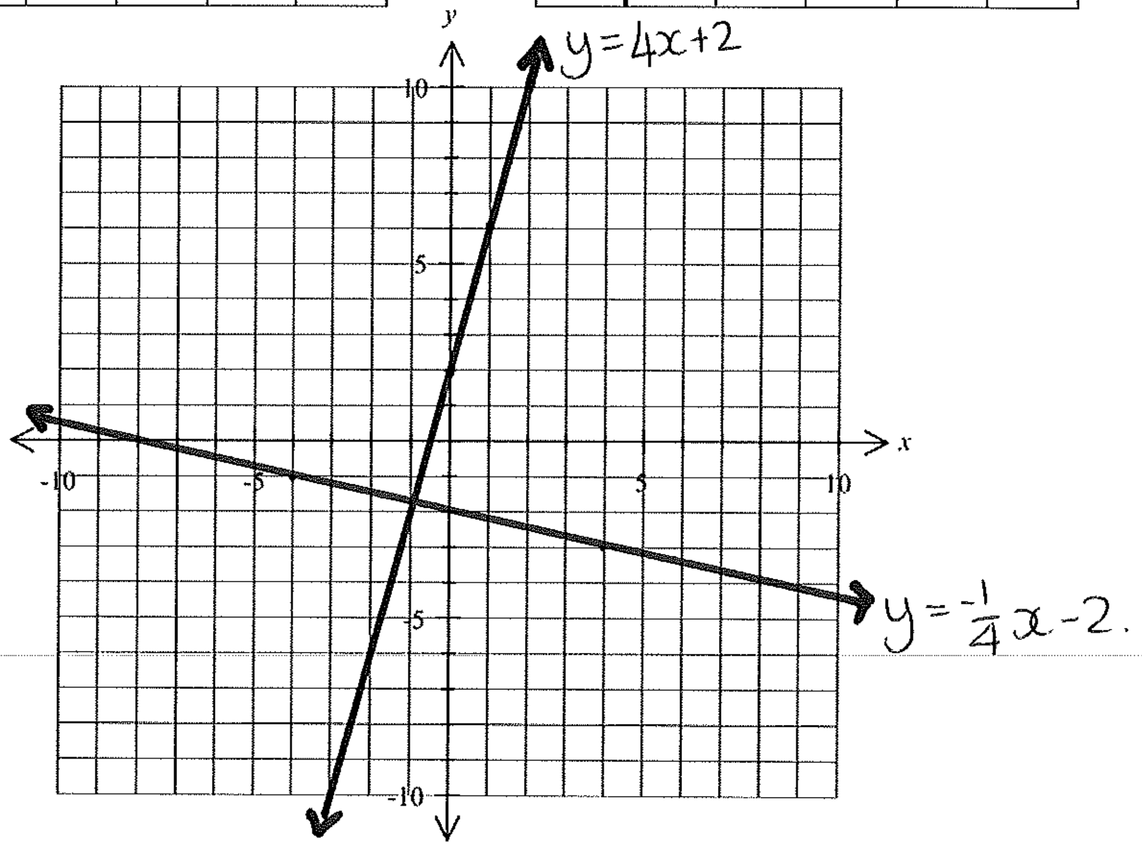
8. For each equation given below, complete the table of values and graph the functions on the axes below.

a) $y = 4x + 2$

x	−2	−1	0	1	2
y	-6	-2	2	6	10

b) $y = -\frac{1}{4}x - 2$

x	−4	−2	0	2	4
y	-1	-1.5	-2	-2.5	-3



9. Measure the angle between the two graphs. What do you notice? What is this relationship called?

Angle in between is 90°.

Perpendicular lines.

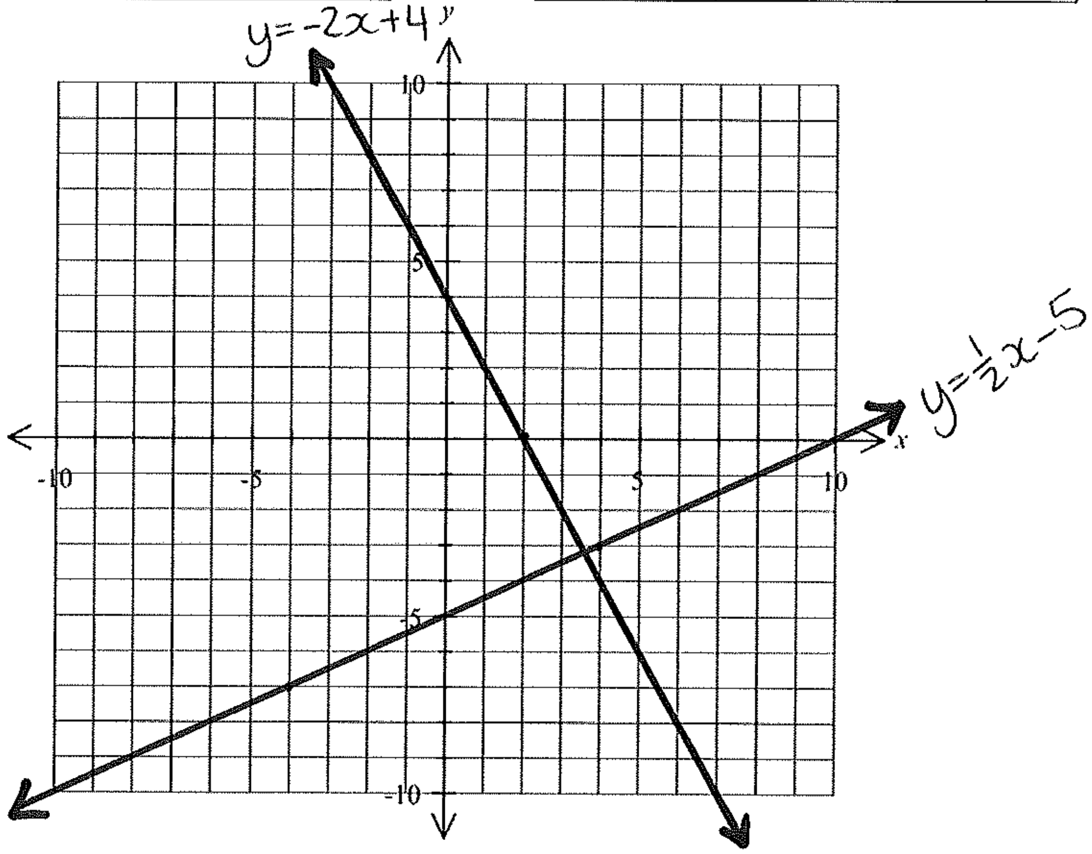
10. For each equation given below, complete the table of values and graph the functions on the axes below.

a) $y = -2x + 4$

x	-2	-1	0	1	2
y	8	6	4	2	0

b) $y = \frac{1}{2}x - 5$

x	-4	-2	0	2	4
y	-7	-6	-5	-4	-3



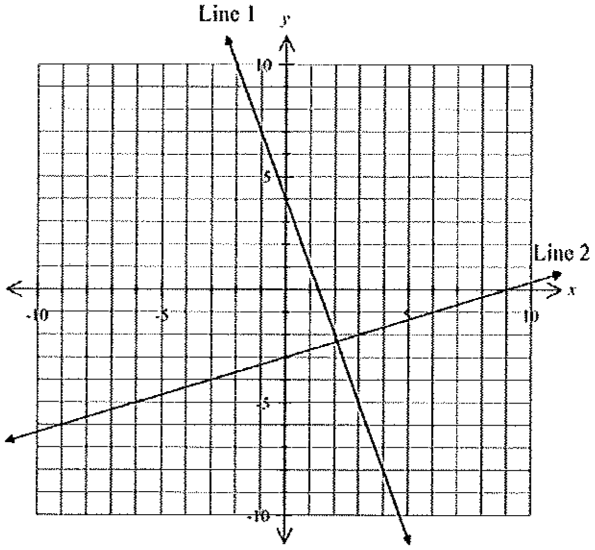
11. Measure the angle between the two graphs. What do you notice? What is this relationship called?

Angle in between is 90°.

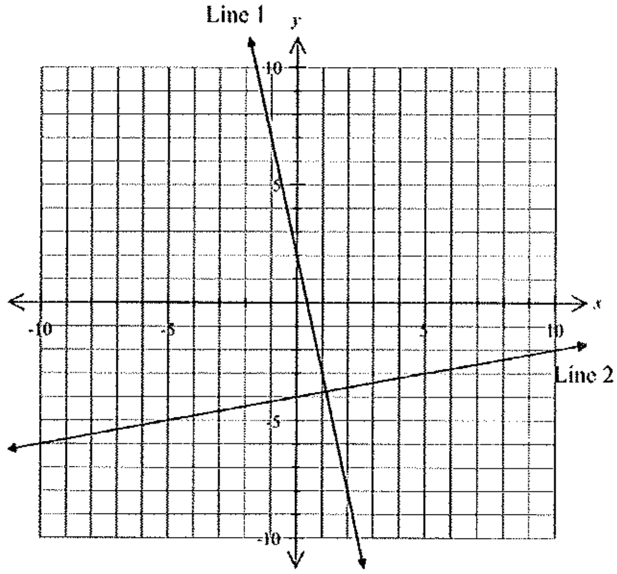
Perpendicular lines.

12. Each graph below contains pairs of lines. Calculate the gradient of each line and record the results in the table below.

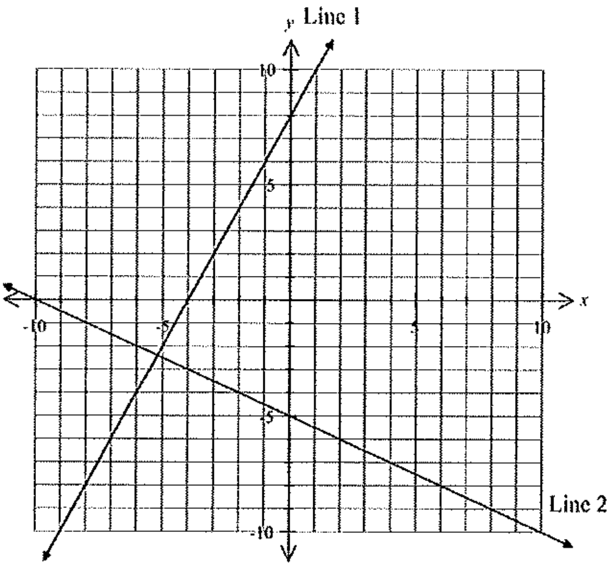
a)



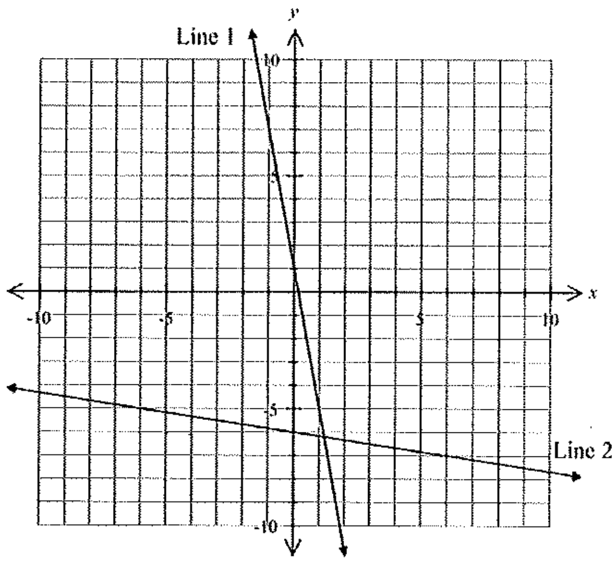
b)



c)



d)



	Line 1 gradient	Line 2 gradient	Line 1 gradient × Line 2 gradient
a)	-3	$\frac{1}{3}$	-1
b)	-5	$\frac{1}{5}$	-1
c)	2	$-\frac{1}{2}$	-1
d)	-6	$-\frac{1}{6}$	1

13. Multiply the two gradients together to fill in the last column of the table.

14. What pattern do you notice about the gradients?

a, b, c multiply to -1

d multiplies to 1.

a, b, c are perpendicular lines, d isn't.

15. Write a conjecture (guess) about the relationship between the gradients of perpendicular lines.

For two lines to be perpendicular, the gradients must multiply to equal -1.

16. Use your conjecture to state whether each pair of linear functions below are perpendicular:

a) $y=5x+5$ and $y=-5x+2$ NO

b) $y=3x-4$ and $y=-\frac{1}{3}x+7$ yes

c) $y=-6x-2$ and $y=-\frac{1}{6}x+9$ NO

d) $y=\frac{2}{3}x+7$ and $y=-\frac{3}{2}x+9$ yes.